Mirpur University of Science and technology

Department of software engineering

Submitted to:

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Subject:

Web Engineering

Submitted by:

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Project Title:

Full-Stack Student Complaint Management System.

1.Introduction:

The Student Complaint Management System (SCMS) is a full-stack web application designed to provide students with a digital platform to submit, track, and manage complaints related to academics, facilities, or administrative issues.

The system also provides an administrative panel where administrators can view, update, and manage complaints efficiently.

This project simulates a real-world web development workflow by integrating frontend design, backend logic, and a relational database system.

2. Problem Statement

Traditional complaint handling systems are often manual and inefficient. Common issues include:

- Complaints getting lost
- No proper tracking system
- Delayed responses
- Lack of transparency
- No centralized database

To solve these problems, a centralized web-based complaint management system was developed that ensures proper complaint tracking, transparency, and efficiency.

3. System Features

Student Features

- Student Registration
- Secure Login
- Submit Complaint
- Edit Complaint (only if status is Pending)
- Delete Complaint

- View Complaint History
- Reset Password
- Delete Account

Admin Features

- Role-Based Secure Login
- View All Complaints
- Update Complaint Status (Pending / In Progress / Resolved)
- Dashboard Statistics (Total, Resolved, Active Students)
- Change Password

4. Technology Stack

Frontend

- HTML5
- CSS3
- JavaScript

Backend

- Node.js
- Express.js

Database

- MySQL (Relational Database)

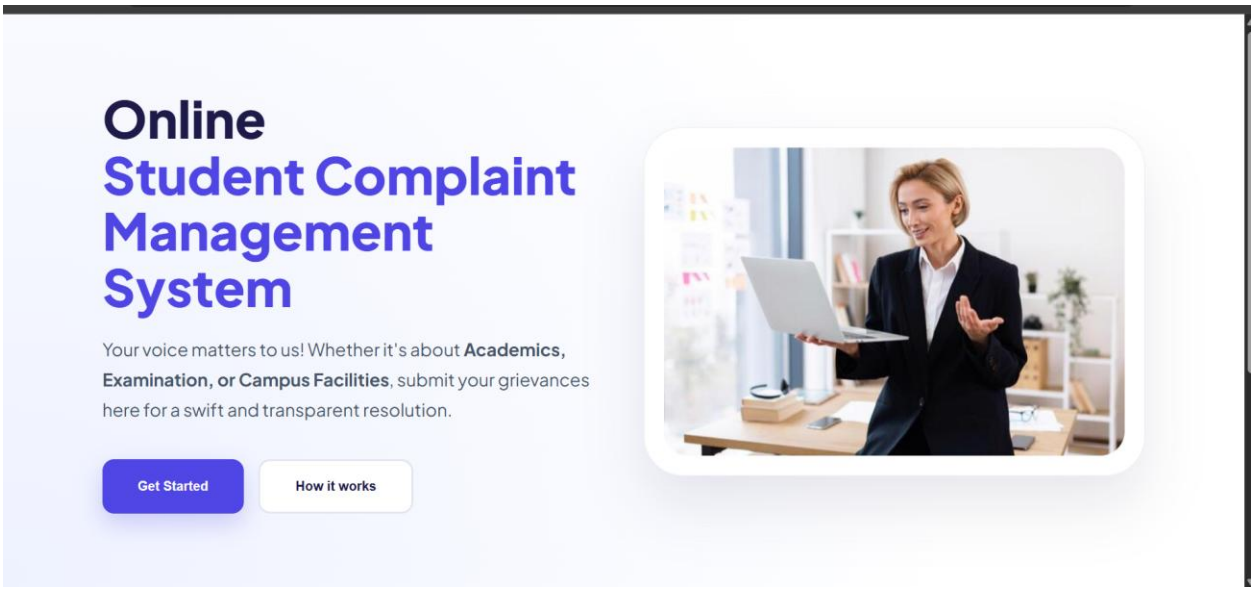
Version Control

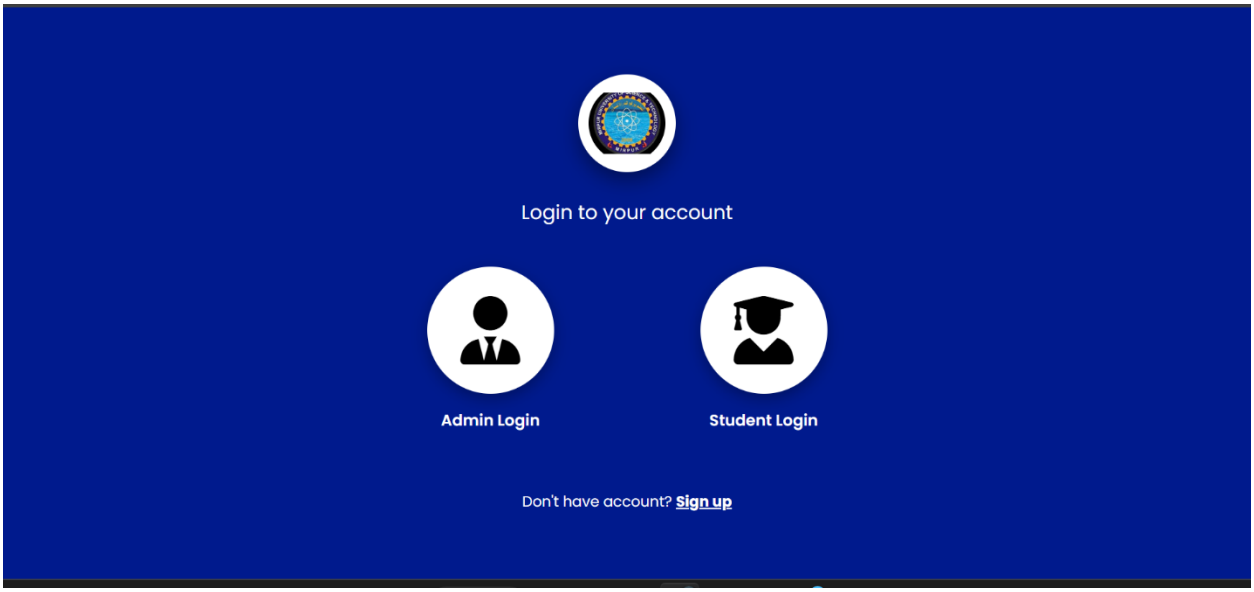
- Git
- GitHub

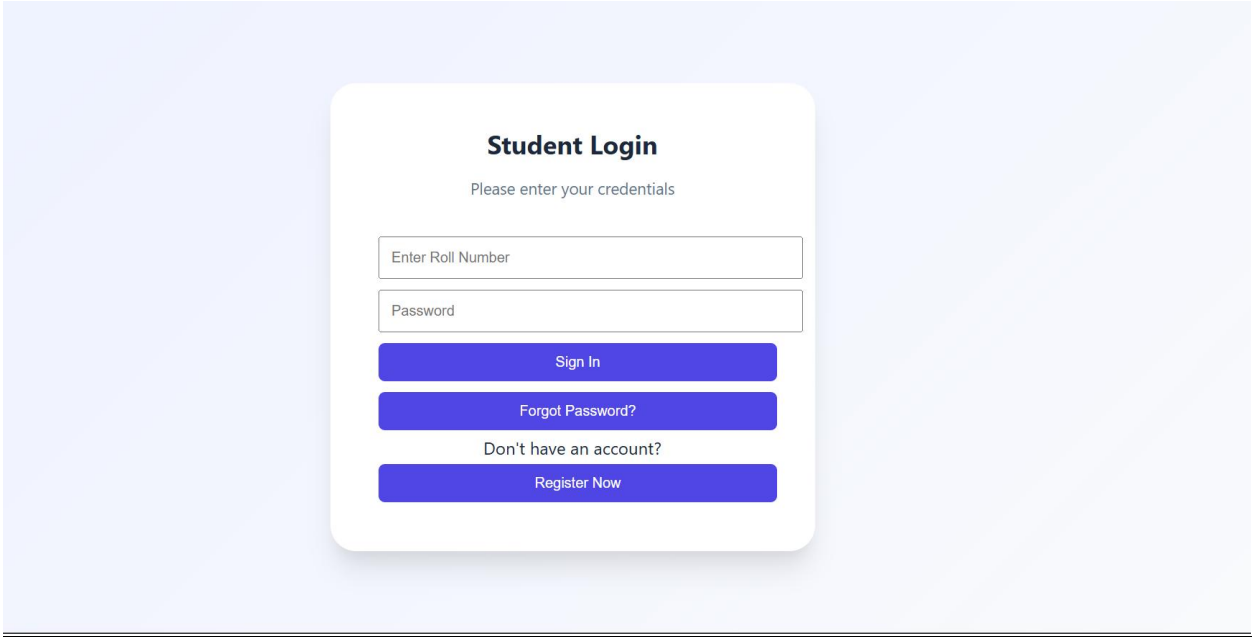
5. Frontend Design:

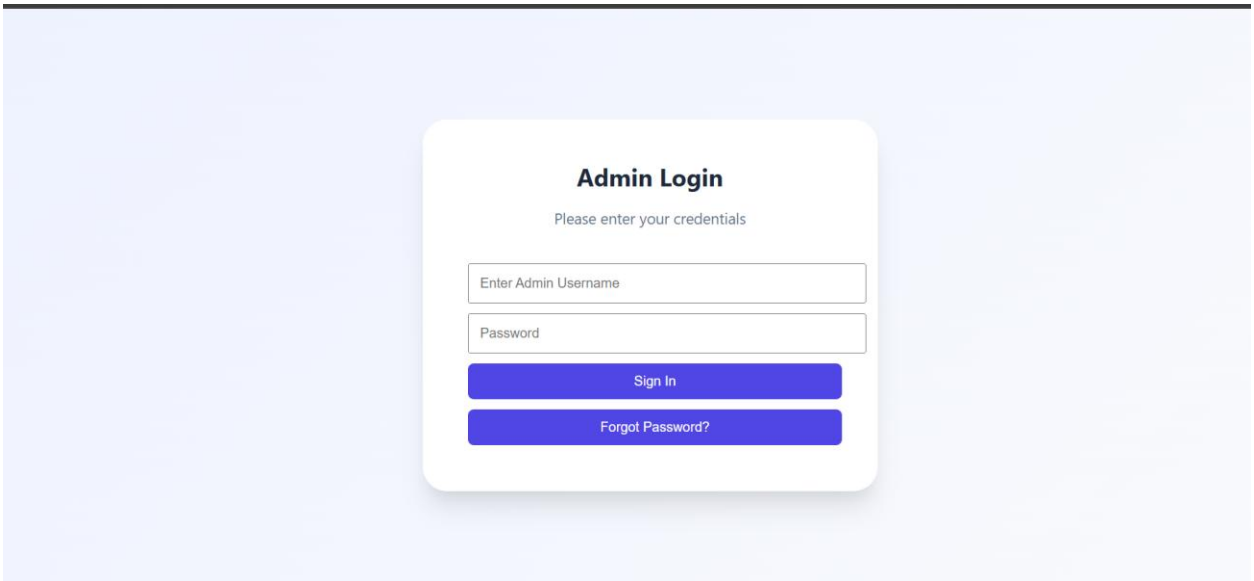
The frontend consists of multiple well-designed pages:

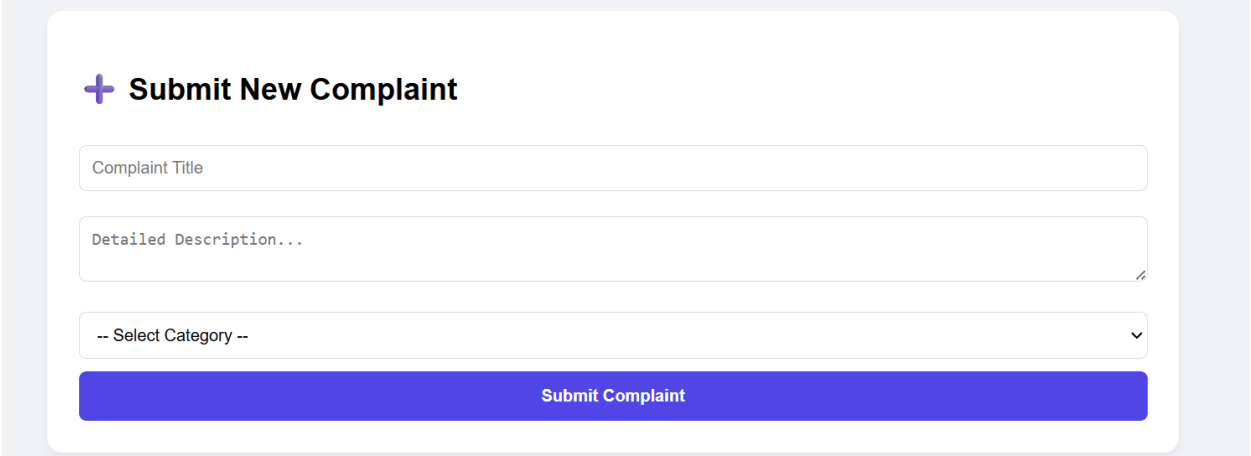
- **Login Page**
- **Registration Page**
- **Student Dashboard**
- **Admin Dashboard**
- **Complaint Form Page**





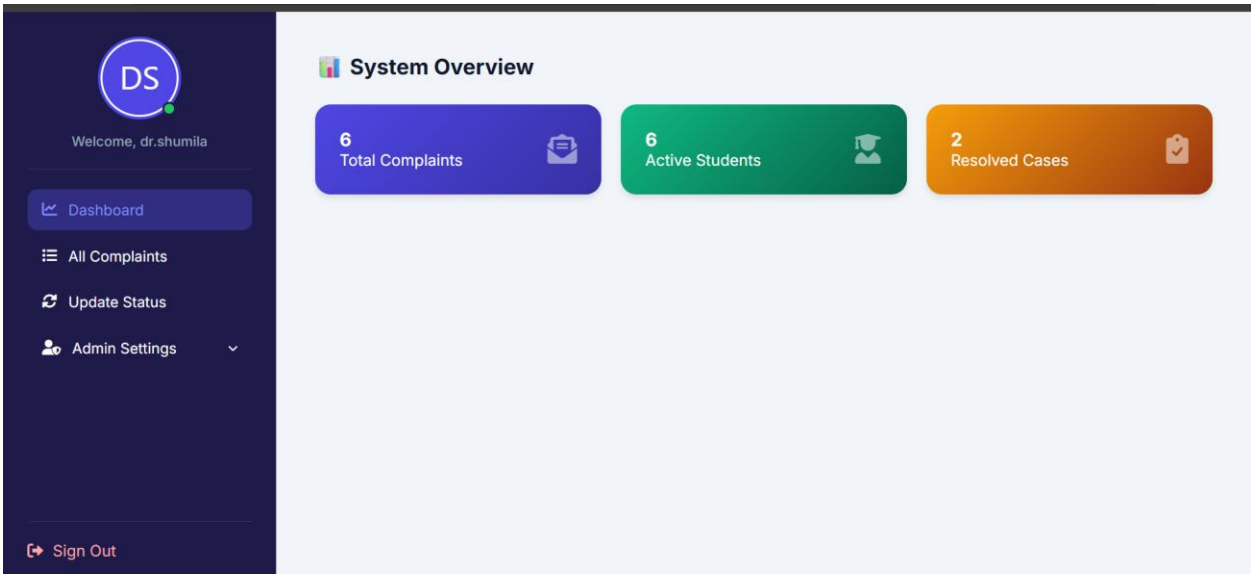

Student Login
Please enter your credentials
Enter Roll Number
Password
Sign In
Forgot Password?
Don't have an account?
Register Now


Admin Login
Please enter your credentials
Enter Admin Username
Password
Sign In
Forgot Password?



UI/UX Design

- Responsive layout
- Sidebar navigation
- Dashboard statistics cards
- CSS styling using Flexbox
- Interactive buttons and status badges



Client-Side Validation

JavaScript validation is implemented to ensure correct input before sending data to the server.

Examples include:

- Required field validation
- Password match validation
- Empty input checks

Example Code:

```
if (!oldPassword || !newPassword) {
  alert("All fields are required.");
}
```

Dynamic Data Rendering

The frontend uses the Fetch API to retrieve data from the backend:

```
const res = await fetch("/admin-complaints");
const data = await res.json();
```

Data is dynamically displayed using DOM manipulation.

6. Backend Architecture:

The backend is developed using Node.js and Express.js and follows RESTful API principles.

System Architecture Flow

Frontend → REST API → Express Server → MySQL Database → Response → Frontend Display

Routing Structure (CRUD Implementation)

Method	Route	Operation
POST	/login	Login
POST	/signup	Create User
POST	/complaint	Create Complaint
GET	/student-complaints/:id	Read Complaints
PUT	/edit-complaint/:id	Update Complaint
DELETE	/delete-complaint/:id	Delete Complaint
PUT	/status/:id	Update Status

7. Database Design:

1. Users Table

Field	Type
UserID (Primary Key)	INT
Username	VARCHAR
Password	VARCHAR
Role	VARCHAR
Session	VARCHAR

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

UserID	Username	Password	Role	Session
1	dr.shumila	shumila@1234	Admin	NULL
2	FA23-BSE-053	nayab@123	Student	2022-26
3	FA23-BSE-029	hira@123	Student	2023-27
4	FA23-BSE-034	iman@1234	Student	2022-26
6	FA23-BSE-030	hira@123	Student	2022-26
7	FA23-BSE-070	aman2123	Student	2023-27
NULL	NULL	NULL	NULL	NULL

users 39 | Apply

Complaints Table

Field	Type
ComplaintID (Primary Key)	INT
Title	VARCHAR
Description	TEXT
Category	VARCHAR
Status	VARCHAR
SubmittedBy (Foreign Key)	INT

ComplaintID	Title	Description	Category	Status	SubmittedB
1	attandance issue	i attend class but teacher mark me absent.i ask...	Facility	Resolved	3
3	marks issue	economic marks issue.teacher not recheck pape...	Examination	Resolved	6
4	Library Books Not Available	The latest edition of Engineering Mathematics is...	Academic	Pending	2
5	Result Error in Semester 3	My CGPA is showing as 0.0 despite passing all s...	Academic	Pending	NULL
6	Portal Login Lag	The student portal takes too much time to load ...	Academic	Pending	7
7	ID Card Delay	I applied for a duplicate ID card two weeks ago ...	Academic	Pending	4
NULL	NULL	NULL	NULL	NULL	NULL

Relationship

- One User can submit multiple complaints.
- SubmittedBy is a Foreign Key referencing UserID in the Users table.

Data Integrity

- Primary Keys ensure uniqueness
- Foreign Keys maintain relationships
- Status-based editing restrictions
- Role-based access control

8. GitHub Repository

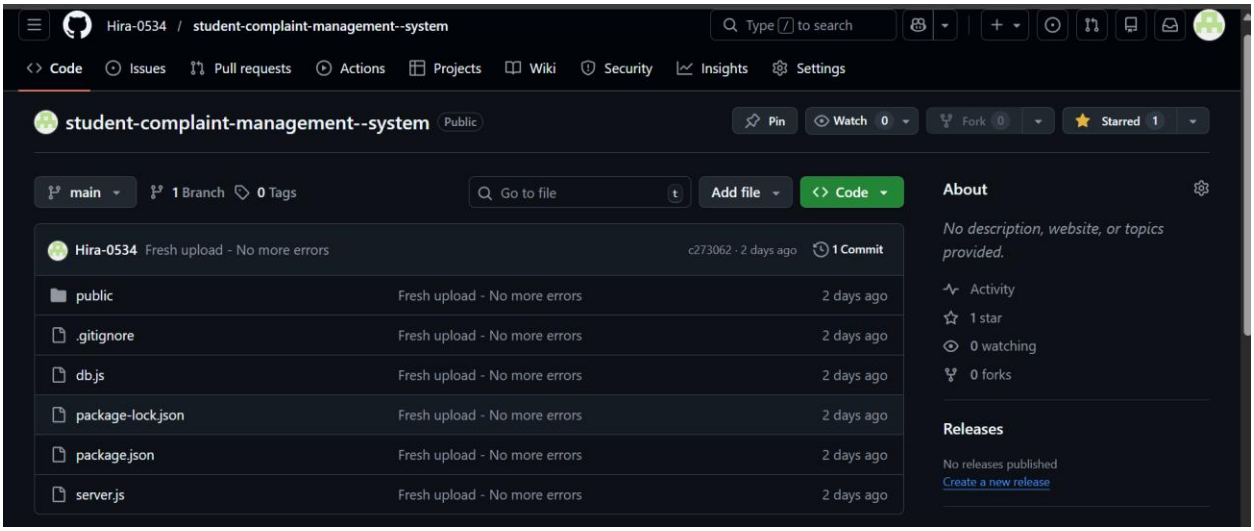
The project is uploaded to a public GitHub repository and includes:

- Complete source code
- Organized folder structure
- Database connection files
- README.md
- Setup instructions
- Project report

The README includes:

- Project description
- Technology stack
- Installation steps
- Execution instructions

Link: https://github.com/Hira-0534/student-complaint-management--system



9.System Workflow

Student Workflow

1. Register an account
2. Login
3. Submit complaint
4. View complaint status
5. Edit or delete if pending

Admin Workflow

1. Login
2. View dashboard
3. Update complaint status
4. Manage complaints

10.Conclusion & Future Enhancements

Conclusion:

The Student Complaint Management System successfully demonstrates full-stack web development using frontend technologies, backend server logic, and relational database integration.

The system implements complete CRUD operations, role-based authentication, and proper data management techniques.

This project reflects real-world application architecture and follows structured development practices.

Future Enhancements:

- Password hashing for improved security
- Email notifications for status updates
- File attachment support
- Advanced filtering and reporting
- Staff role integration
- Adding a 3rd table for **Categories** to further normalize the database and adding automated email alerts.